The documentation and process conversion measures necessary to comply with this document shall be completed by 28 February 2019.

INCH POUND

MIL-PRF-19500/181K
28 November 2018
SUPERSEDING
MIL-PRF-19500/181J
16 February 2011

PERFORMANCE SPECIFICATION SHEET

# SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER, TYPES 2N718A, 2N1613, AND 2N1613L, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

## 1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for NPN silicon, low-power transistors. Three levels of product assurance are provided (JAN, JANTX, and JANTXV) for each device type as specified in MIL-PRF-19500.

* 1.2 Physical dimensions. The device packages for the encapsulated device types are as follows: (2N718A) (TO-18) in accordance with figure 1, and (2N1613 and 2N1613L) (similar to TO-39 and TO-5) in accordance with figure 2.

1.3 Maximum ratings. Unless otherwise specified, $T_C$ = +25°C.

| Type | $P_T$ (1) $T_C$ = +25°C | $P_T$ (2) $T_A$ = +25°C | $V_{CBO}$ | $V_{CEO}$ | $V_{EBO}$ | $I_C$ | $V_{CER}$ $R_{BE}$ = 10 Ω | $R_{\theta JC}$ | $R_{\theta JA}$ | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| | W | W | V dc | V dc | V dc | mA dc | V dc | °C/W | °C/W | °C |
| 2N718A | 1.8 | 0.5 | 75 | 30 | 7 | 500 | 50 | 97 | 325 | -65 |
| 2N1613 | 3.0 | 0.8 | 75 | 30 | 7 | 500 | 50 | 58 | 175 | to |
| 2N1613L | 3.0 | 0.8 | 75 | 30 | 7 | 500 | 50 | 58 | 175 | +200 |

(1) Derate linearly at 17.2 mW/°C for type 2N1613 and 2N1613L and at 10.3 mW/°C for type 2N718A for $T_C$ > +25°C.
(2) See figures 3 and 4.

* Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.



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1.4 Primary electrical characteristics.

| Limits | $h_{FE1}$ (1) | $h_{FE2}$ (1) | $h_{FE3}$ (1) | $h_{FE4}$ (1) | $V_{CE(SAT)}$ (1) | $\|h_{FE}\|$ at 20 MHz |
|---|---|---|---|---|---|---|
| | $V_{CE}$ = 10 V dc<br>$I_C$ = 0.1 mA dc | $V_{CE}$ = 10 V dc<br>$I_C$ = 10 mA dc | $V_{CE}$ = 10 V dc<br>$I_C$ = 150 mA dc | $V_{CE}$ = 10 V dc<br>$I_C$ = 500 mA dc | $I_C$ = 150 mA dc<br>$I_B$ = 15 mA dc | $V_{CE}$ = 10 V dc<br>$I_C$ = 50 mA dc<br>f = 20 MHz |
| Min<br>Max | 20 | 35 | 40<br>120 | 20 | V dc<br>1.5 | 3 |

(1) Pulsed (see 4.5.1).

* 1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

* 1.5.1 JAN certification mark and quality level.

* 1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", and "JANTXV".

* 1.5.2 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

* 1.5.2.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

* 1.5.2.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "718" and “1613".

* 1.5.3 Suffix symbols. The following suffix letters are incorporated in the PIN in the order listed in the table as applicable:

| | |
|---|---|
| A | Indicate a modified version of the base part number. |
| L | For L suffix devices, dimension LL = 1.50 inches (38.10 mm) min. and 1.75 inches (44.45 mm) max. |

* 1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.



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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at https://quicksearch.dla.mil.

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

* 3.4 Interface and physical dimensions. The design, construction and physical dimensions shall be as specified in MIL-PRF-19500 and figure 1 (TO18) and figure 2 (Similar to TO-39 and TO-5) herein.

3.4.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

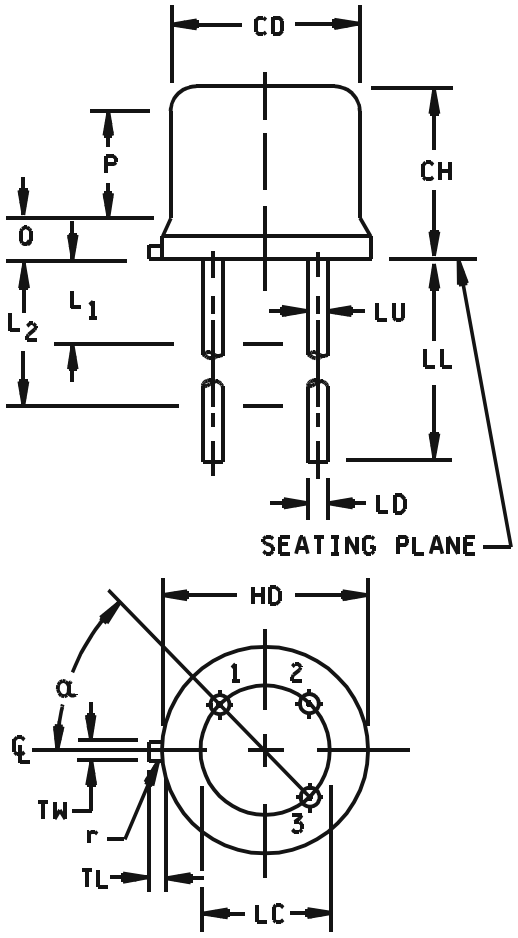
3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.



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| Symbol | Inches | | Millimeters | | Notes |
|---|---|---|---|---|---|
| | Min | Max | Min | Max | |
| CD | .178 | .195 | 4.52 | 4.95 | |
| CH | .170 | .210 | 4.32 | 5.33 | |
| HD | .209 | .230 | 5.31 | 5.84 | |
| LC | .100 TP | | 2.54 TP | | 5 |
| LD | .016 | .021 | 0.41 | 0.53 | 8, 9 |
| LL | .500 | .750 | 12.70 | 19.05 | 7, 9 |
| LU | .016 | .019 | 0.41 | 0.48 | 4, 8, 9 |
| L1 | | .050 | | 1.27 | 9 |
| L2 | .250 | | 6.35 | | 9 |
| TL | .028 | .048 | 0.71 | 1.22 | 5 |
| TW | .036 | .046 | .91 | 1.17 | |
| P | .100 | | 2.54 | | 3 |
| Q | | .030 | | 0.76 | 6 |
| r | | .010 | | .025 | |
| α | 45° TP | | 45° TP | | |



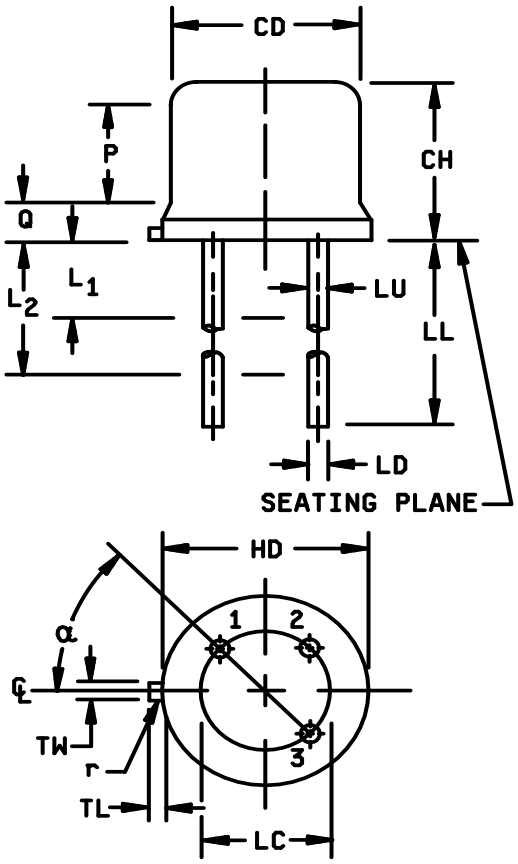

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. This zone is controlled for automatic handling. The variation in actual diameter within this zone shall not exceed .010 inch (0.254 mm).
4. (Three leads) LU applies between $L_1$ and $L_2$. LD applies between $L_2$ and .5 inch (12.70 mm) from seating plane. Diameter is uncontrolled in $L_1$ and beyond .5 inch (12.70 mm) from seating plane.
5. Measured from maximum diameter of the actual device.
6. Details of outline in this zone optional.
7. The collector shall be electrically connected to the case.
8. Lead number 1 - emitter; lead number 2 - base; lead number 3 - collector.
9. All three leads.
10. In accordance with ANSI Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 1. Physical dimensions 2N718A (TO-18).



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| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeter | | |
| | Min | Max | Min | Max | |
| CH | .240 | .260 | 6.10 | 6.60 | |
| LC | .200 TP | | 5.08 TP | | 7 |
| LD | .016 | .021 | 0.41 | 0.53 | 8,9 |
| LL | See notes 12, and 13 | | | | |
| LU | .016 | .019 | 0.41 | 0.48 | 8,9 |
| $L_1$ | | .050 | | 1.27 | 8, 9 |
| $L_2$ | .250 | | 6.35 | | 8, 9 |
| HD | .335 | .370 | 8.51 | 9.40 | |
| CD | .305 | .335 | 7.75 | 8.51 | |
| P | .100 | | 2.54 | | 6 |
| Q | | .050 | | 1.27 | 5 |
| r | | .010 | | 0.25 | |
| TL | .029 | .045 | 0.74 | 1.14 | 4 |
| TW | .028 | .034 | 0.71 | 0.86 | 0.71 |
| $\alpha$ | 45° TP | | 45° TP | | 7 |




NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of 0.011 inch (0.28 mm).
4. TL measured from maximum HD.
5. Outline in this zone is not controlled.
6. CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane .054 +.001, -.000 inch (1.37 +0.03, -0.000 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at a maximum material condition (MMC) relative to the tab at MMC. The device may be measured by direct methods or by the gauging procedure.
8. LU applies between $L_1$ and $L_2$. LU applies between $L_2$ and LL minimum. Diameter is uncontrolled in $L_1$ and beyond LL minimum.
9. All three leads.
10. The collector shall be electrically and mechanically connected to the case.
11. r (radius) applies to both inside corners of tab.
12. For transistor types 2N1613, dimension LL is .500 inch (12.70 mm) minimum, and .750 inch (19.05 mm) maximum.
13. For transistor types 2N1613L, dimension LL is 1.500 inches (38.10 mm) minimum, and 1.750 inches (44.45 mm) maximum.
14. Lead number 1 - emitter; lead number 2 - base; lead number 3 - collector.

FIGURE 2. Physical dimensions 2N1613 and 2N1613L (similar to TO-5 and TO-39).



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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen | Measurement |
|---|---|
| | JANTX and JANTXV levels |
| (1) 3c | Thermal impedance see 4.3.2 |
| 9 | Not applicable |
| 11 | $I_{CBO2}$, $h_{FE3}$ |
| 12 | See 4.3.1 |
| 13 | Subgroup 2 of table I herein; $\Delta I_{CBO2}$ = 100 percent of initial value or 5 nA dc, whichever is greater; $\Delta h_{FE3}$ = $\pm$15 percent |

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: $V_{CB}$ = 10 - 30 V dc. Power shall be applied to achieve $T_J$ = +135°C minimum using a minimum $P_D$ = 75 percent of $P_T$ maximum rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, $T_J$, and mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

* 4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$, $t_{MD}$, ($V_C$ and $V_H$ where appropriate). See table II, group E, subgroup 4 herein.



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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table E-VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in 4.4.2.1 for JAN, JANTX, and JANTXV group B testing. Delta requirements for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.1 and shall be in accordance with 4.5.2 herein.

4.4.2.1 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new “assembly lot” option is exercised, the failed assembly lot shall be scrapped.

| Step | Method | Condition |
|---|---|---|
| 1 | 1026 | Steady-state life: 1,000 hours minimum, $V_{CB}$ = 10 V dc, power shall be applied to achieve $T_J$ = +150°C minimum using a minimum of $P_D$ = 75 percent of maximum rated $P_T$ as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. |
| 2 | 1048 | Blocking life, $T_A$ = 150°C, $V_{CB}$ = 80 percent of rated voltage, without going over the maximum rated $V_{CE}$, 48 hours minimum. n = 45 devices, c = 0. |
| 3 | 1032 | High-temperature life (non-operating), t = 340 hours, $T_A$ = +200°C. n = 22, c = 0. |

4.4.2.2 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. See MIL-PRF-19500.

b. Shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

* 4.4.3 Group C inspection, Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and 4.4.3.1 herein for group C testing.



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4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| C2 | 2036 | Test condition E. |
| C5 | 3131 | $R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3) and in accordance with thermal impedance curves. |
| C6 | | Not applicable. |

* 4.4.4 Group E Inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500 and as specified in table II herein.

4.5 Method of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Delta requirements. Delta requirements shall be as specified below:

| Step | Inspection | MIL-STD-750 | | Symbol | Limit | Unit |
|---|---|---|---|---|---|---|
| | | Method | Conditions | | | |
| 1 | Collector-base cutoff current | 3036 | Bias condition D, $V_{CB}$ = 60 V dc | $\Delta I_{CB02}$ (1) | 100 percent of initial value or 5 nA dc, whichever is greater. | |
| 2 | Forward current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 150 mA dc; pulsed see 4.5.1 | $\Delta h_{FE3}$ (1) | ±25 percent change from initial reading. | |

(1) Devices which exceed the group A limits for this test shall not be accepted.



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* TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 2/ | | | | | | |
| Visual and mechanical 3/ examination | 2071 | n = 45 devices, c = 0 | | | | |
| Solderability 3/ | 2026 | n = 15 leads, c = 0 | | | | |
| Resistance to solvents 3/ 4/ | 1022 | n = 15 devices, c = 0 | | | | |
| * Salt atmosphere (corrosion) 3/ | 1041 | n = 6 devices, c = 0, (For laser marked devices only) | | | | |
| Temp cycling 3/ | 1051 | Test condition C, 25 cycles. n = 22 devices, c = 0 | | | | |
| Hermetic seal<br>Fine leak<br>Gross leak | 1071 | n = 22 devices, c = 0 | | | | |
| Electrical measurements | | Group A, subgroup 2 | | | | |
| Bond strength 3/ | 2037 | Precondition $T_A$ = +250°C at t = 24 hrs or $T_A$ = +300°C at t = 2 hrs n = 11 wires, c = 0 | | | | |
| Decap internal visual | 2075 | n = 4 device, c = 0 | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance 5/ | 3131 | See 4.3.2 | $Z_{\theta JX}$ | | | °C/W |
| Collector-to-base cutoff current | 3036 | Bias condition D, $V_{CBO}$ = 75 V dc pulsed (see 4.5.1) | $I_{CBO1}$ | | 10 | µA dc |
| Emitter to base cutoff current | 3061 | Bias condition D; $V_{EBO}$ = 7 V dc pulsed (see 4.5.1) | $I_{EBO1}$ | | 10 | µA dc |
| Breakdown voltage, collector-emitter | 3011 | Bias condition D; $I_C$ = 100 µA dc; pulsed (see 4.5.1) | $V_{(BR)CEO}$ | 30 | | V dc |
| * Breakdown voltage collector-emitter | 3011 | Bias condition B, $I_C$ = 100 µA dc, pulsed (see 4.5.1), $R_{BE}$ = 10 Ω | $V_{(BR)CER}$ | 50 | | V dc |
| Collector to base cutoff current | 3036 | Bias condition D, $V_{CB}$ = 60 V dc | $I_{CBO2}$ | | 10 | nA dc |
| Emitter-base cutoff current | 3061 | Bias condition D, $V_{EB}$ = 5 V dc | $I_{EBO2}$ | | 10 | nA dc |
| Collector-emitter saturated voltage | 3071 | $I_C$ = 150 mA dc, $I_B$ = 15 mA dc, pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | 1.5 | V dc |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Base-emitter saturated voltage | 3066 | Test condition A, $I_C$ = 150 mA dc, $I_B$ = 15 mA dc, pulsed (see 4.5.1) | $V_{BE(sat)1}$ | | 1.3 | V dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc, $I_C$ = 0.1 mA dc, pulsed (see 4.5.1) | $h_{FE1}$ | 20 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc, $I_C$ = 10 mA dc, pulsed (see 4.5.1) | $h_{FE2}$ | 35 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 150 mA dc pulsed (see 4.5.1) | $h_{FE3}$ | 40 | 120 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc; $I_C$ = 500 mA dc, pulsed (see 4.5.1) | $h_{FE4}$ | 20 | | |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +150°C | | | | |
| Collector to base cutoff current | 3036 | Bias condition D, $V_{CB}$ = 60 V dc | $I_{CBO3}$ | | 10 | µA dc |
| Low temperature operation | | $T_A$ = -55C | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 10 V dc, $I_C$ = 10 mA dc, pulsed (see 4.5.1) | $h_{FE5}$ | 20 | | |
| Subgroup 4 | | | | | | |
| Small signal short circuit forward current transfer ratio | 3206 | $V_{CE}$ = 5 V dc, f = 1 kHz, $I_C$ = 1 mA dc: | $h_{fe}$ | 30 | 100 | |
| | | $V_{CE}$ = 10 V dc, f = 1 kHz, $I_C$ = 5 mA dc | $h_{fe}$ | 35 | 150 | |
| Magnitude of common emitter small-signal short-circuit forward-current transfer ratio | 3306 | $V_{CE}$ = 10 V dc, f = 20 MHz, $I_C$ = 50 mA dc | $\|h_{fe}\|$ | 3 | | |
| Small signal short circuit input impedance | 3201 | $V_{CB}$ = 10 V dc, f = 1 kHz, $I_C$ = 5 mA dc | $h_{ib}$ | 4 | 8 | ohms |
| Small signal open-circuit output admittance | 3216 | $V_{CB}$ = 10 V dc, f = 1 kHz, $I_C$ = 5 mA dc | $h_{ob}$ | 0 | 1.0 | µ ohms |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 – Continued | | | | | | |
| Small signal open circuit reverse voltage transfer ratio | 3211 | $V_{CB}$ = 10 V dc, f = 1 kHz, $I_C$ = 5 mA dc | $h_{rb}$ | | $3 \times 10^{-4}$ | |
| Open circuit output capacitance | 3236 | $V_{CB}$ = 10 V dc, $I_E$ = 0, 100 kHz ≤ f ≤ 1 MHz | $C_{obo}$ | | 25 | pF |
| Pulse response | 3251 | Test condition A, except test circuit and pulse requirements. See figure 5 herein. | $t_{on} + t_{off}$ | | 30 | ns |
| Subgroups 5 and 6 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500.
2/ For resubmission of failed subgroup 1, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for laser marked devices.
5/ This test required for the following end-point measurements only:
* Group B, step 1 (TX and TXV).
Group C, subgroup 2 and 6.



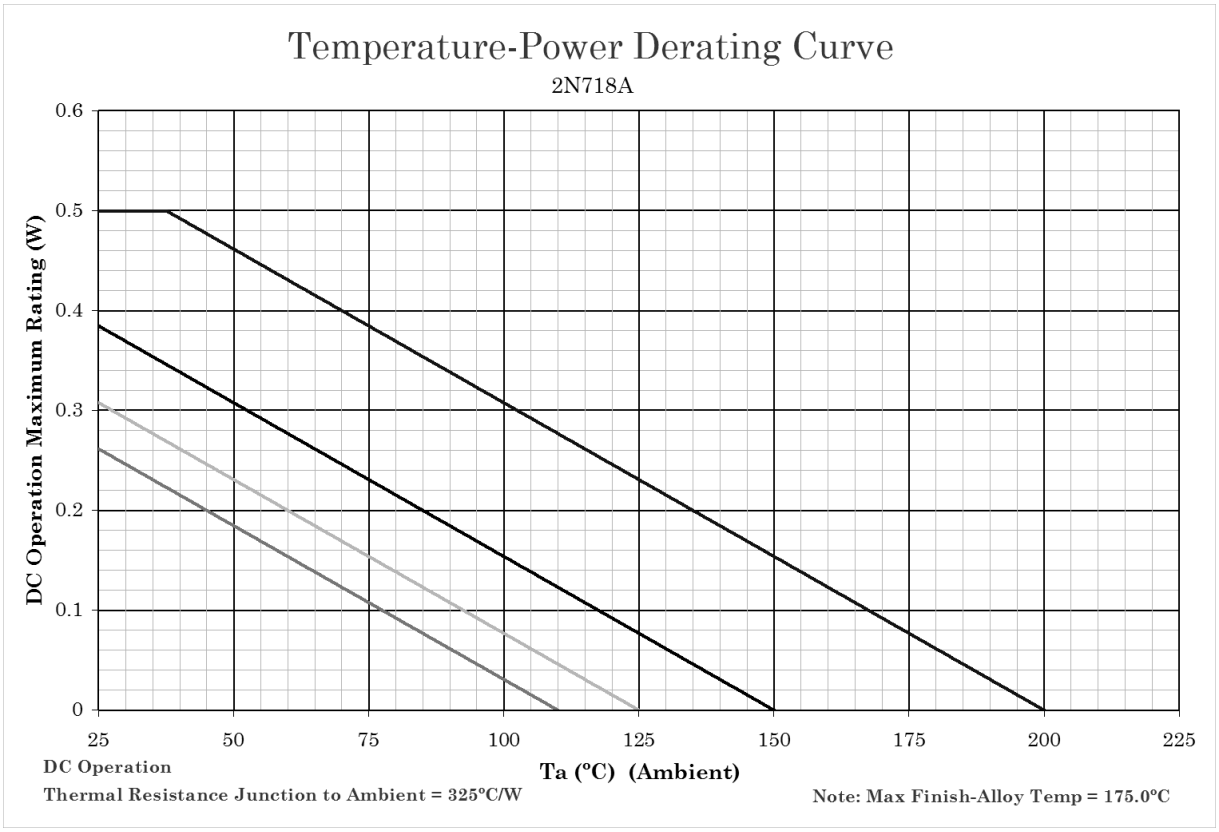
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* TABLE II. Group E inspection (all quality levels) - for qualification or re-qualification only.

| Inspection | MIL-STD-750 | | Qualification |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | |
| * Temperature cycling (air to air) | 1051 | Test condition C, 500 cycles. | 45 devices c = 0 |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See group A, subgroup 2 and 4.5.2 herein. | |
| Subgroup 2 | | | 45 devices c = 0 |
| Intermittent life | 1037 | Intermittent operation life: $V_{CB}$ = 10 V dc, 6,000 cycles, Adjust device current, or power, to achieve a minimum $\Delta T_J$ of +100°C. | |
| Electrical measurements | | See group A, subgroup 2 and 4.5.2 herein. | |
| Subgroup 4 | | | |
| Thermal impedance curves. | | See MIL-PRF-19500. | Sample size N/A |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | 11 devices |
| Electrostatic discharge (ESD) | 1020 | | |
| Subgroup 8 | | | 45 devices c = 0 |
| Reverse stability | 1033 | Condition B. | |



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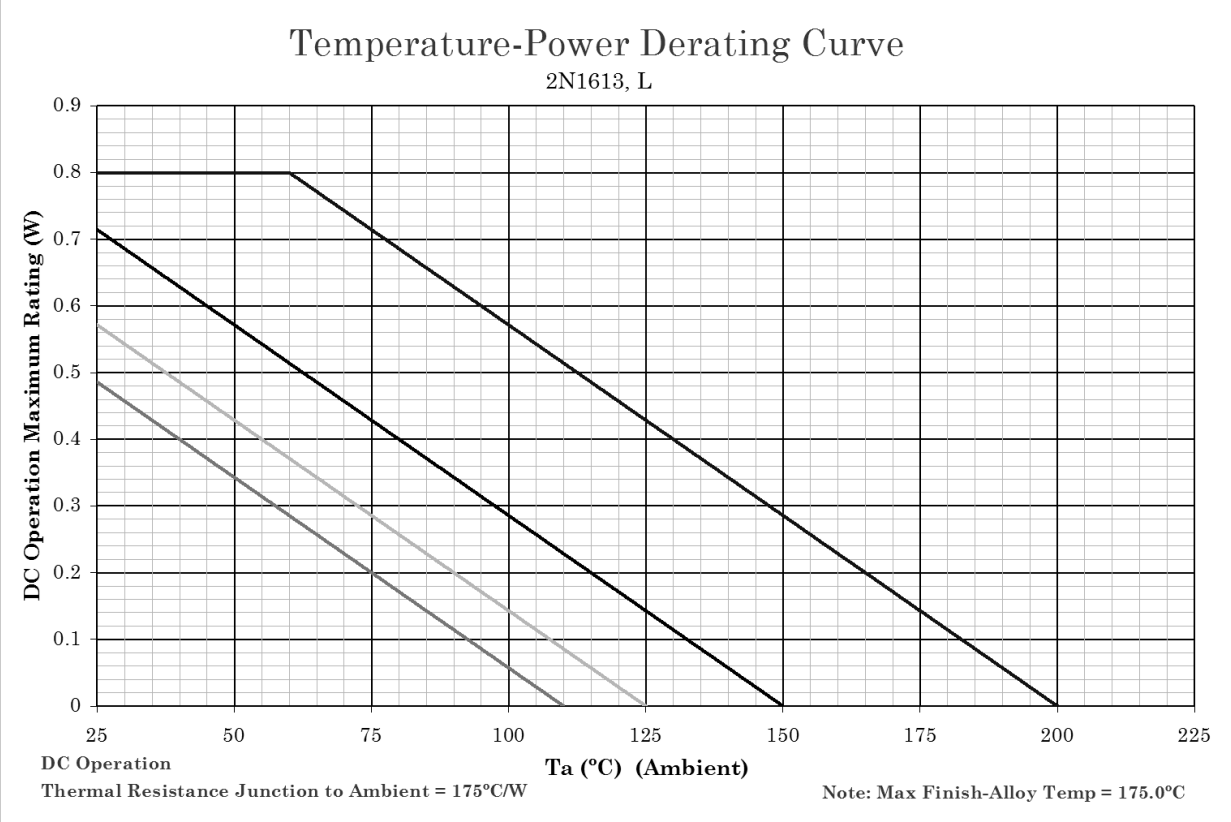
NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J$ ≤, 125°C, and 110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 3. Temperature-power derating for 2N718A (TO-18 package).



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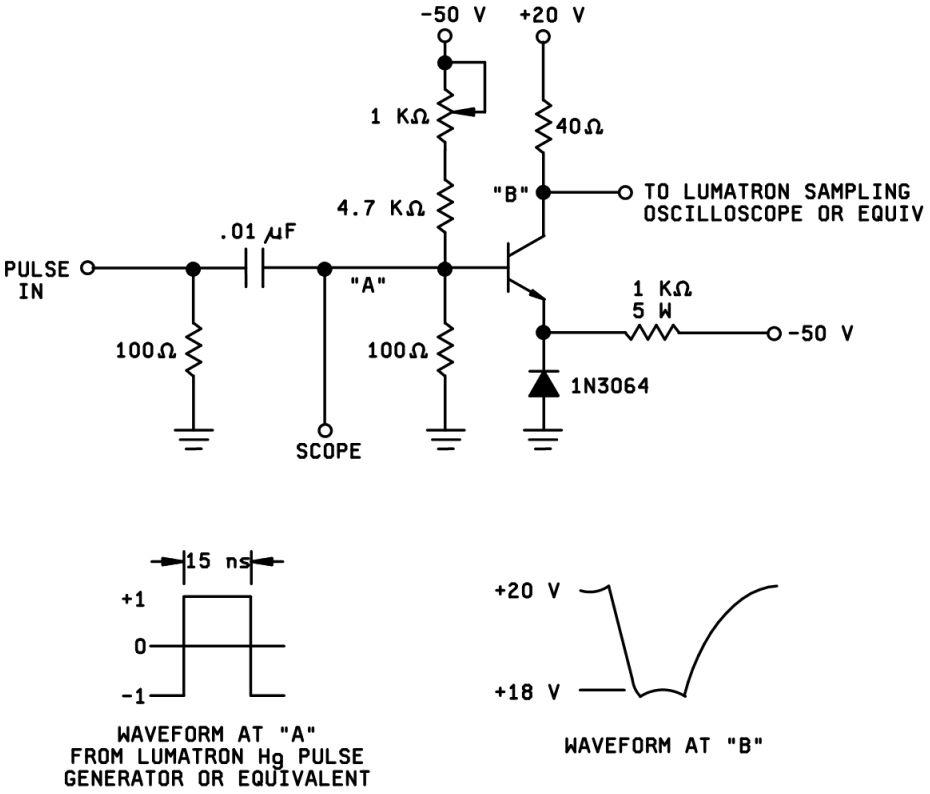

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ $T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum $T_J$ allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J$ ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J$ ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J$ ≤, 125°C, and 110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 4. Temperature-power derating for 2N1613 and 2N1613L (TO-5 and TO-39 package).



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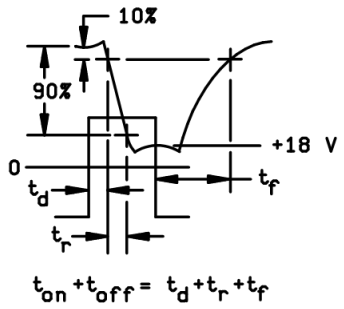




FIGURE 5. Pulse response (turn-on plus turn-off) measurement circuit and waveforms.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

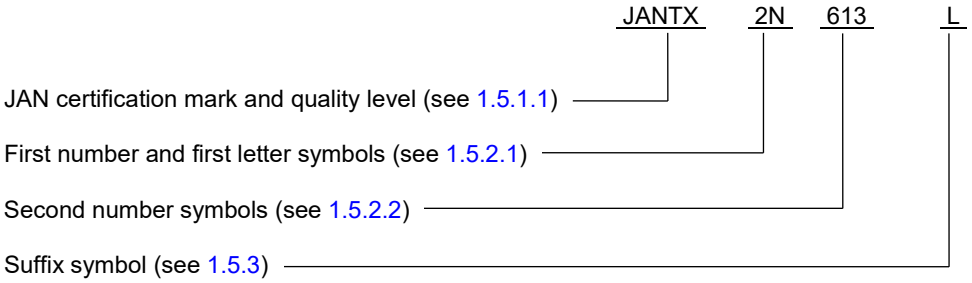
c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

* 6.4 PIN construction example.

* 6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.






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* 6.5 List of PINs.

* 6.5.1 PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

| PINs for type 2N718A, 2N1613, and 2N1613L. | | |
|---|---|---|
| JAN2N718A | JANTX2N718A | JANTXV2N718A |
| JAN2N1613 | JANTX2N1613 | JANTXV2N1613 |
| JAN2N1613L | JANTX2N1613L | JANTXV2N1613L |

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AR, AV, MI, SM
Navy - AS, MC
Air Force - 19, 99

Preparing activity:
DLA - CC

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